

**Department of Computer Science & Engineering (IoT)****Vision of the Department***To be a well-known centre for pursuing computer education through innovative pedagogy, value-based education and industry collaboration.***Mission of the Department***To establish learning ambience for ushering in computer engineering professionals in core and multidisciplinary area by developing Problem-solving skillsthrough emerging technologies.***Session 2026-2027**

Vision: Dream of where you want.	Mission: Means to achieve Vision
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Program Educational Objectives of the program (PEO): (broad statements that describe the professional and career accomplishments)

PEO1	Preparation	P: Preparation	Pep-CL abbreviation pronounce as Pep-si-IL easy to recall
PEO2	Core Competence	E: Environment (Learning Environment)	
PEO3	Breadth	P: Professionalism	
PEO4	Professionalism	C: Core Competence	
PEO5	Learning Environment	L: Breadth (Learning in diverse areas)	

Program Outcomes (PO): (statements that describe what a student should be able to do and know by the end of a program)

Keywords of POs:

Engineering knowledge, Problem analysis, Design/development of solutions, Conduct Investigations of Complex Problems, Engineering Tool Usage, The Engineer and The World, Ethics, Individual and Collaborative Team work, Communication, Project Management and Finance, Life-Long Learning

PSO Keywords: Cutting edge technologies, Research

“I am an engineer, and I know how to apply engineering knowledge to investigate, analyse and design solutions to complex problems using tools for entire world following all ethics in a collaborative way with proper management skills throughout my life.” to contribute to the development of cutting-edge technologies and Research.

Integrity: I will adhere to the Laboratory Code of Conduct and ethics in its entirety.

Sheikh Wasimuddin
Name of Student

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Session	2026-27 (ODD)	Course Name	Lab: Hybrid Computing Systems
Semester	7	Course Code	23IOT1704
Roll No	61	Name of Student	Sheikh Wasimuddin

Practical Number	05
Course Outcome	1. Understand Cloud Computing architecture, service models, deployment models, and virtualization. 2. Apply cloud platforms, storage services, and containerization for application deployment. 3. Implement Edge and Fog Computing solutions for IoT applications. 4. Analyze and develop integrated Cloud-Fog-Edge computing applications based on performance evaluation.
Aim	Containerization and Cloud Deployment using Docker
Theory (100 words)	Containerization is a modern software deployment technique in which an application is packaged together with its libraries, configuration files, runtime environment, and other dependencies. The complete package is called a container. Since all required components are included, the application can operate in a similar manner across different computing environments. Docker is one of the most widely used containerization platforms. It provides tools for creating, running, and managing containers. Docker separates an application from the underlying operating system, making application deployment more portable and efficient. A Docker image is a read-only template containing the application code, runtime, libraries, and configuration required to run an application. A Docker container is an active, isolated environment created from a Docker image. Multiple containers can be created from the same image when required. The configuration for building a Docker image is generally written in a

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	Dockerfile. It contains instructions such as selecting a base image, installing dependencies, copying application files, exposing ports, and specifying the command used to start the application. Docker Deployment Process The typical workflow for containerized application deployment is: Application Source Code → Dockerfile → Docker Image → Docker Container → Cloud Deployment First, the application and its dependencies are prepared. A Dockerfile is then created to describe the application's environment and execution requirements. Docker uses this file to build an image. The image can subsequently be executed as a container on a developer's computer or on a remote server. Cloud Deployment Cloud deployment involves running the containerized application on a cloud infrastructure rather than relying only on a local machine. Cloud services provide computing resources, networking, storage, and other facilities required to make the application available over the Internet. Containerized applications can be deployed using cloud platforms and container orchestration systems. This approach makes it easier to reproduce environments, update applications, allocate resources, and manage multiple instances.
Procedure and Execution (100 Words)	1. Set up Docker: Download and install Docker Desktop, then launch it and ensure the Docker engine is running. 2. Prepare the project: Create a dedicated directory for the application and place the source code, configuration files, and dependency files inside it. 3. Define the container environment: Create a Dockerfile in the project directory and specify the base image, working directory, dependencies, required ports, and application startup command. 4. Open the project terminal: Navigate to the application directory using Command Prompt, PowerShell, or a terminal. 5. Create the Docker image: Execute the Docker build operation to convert the application and Dockerfile instructions into a reusable image.



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6. **Verify the image:** Display the locally available Docker images and confirm that the newly created image is present.
7. **Launch the container:** Create a container from the generated image and map the required application port to the host machine.
8. **Test locally:** Access the application through the mapped port and confirm that all major functions operate properly inside the container.
9. **Prepare the image for sharing:** Assign an appropriate repository and tag to the Docker image so that it can be identified in a container registry.
10. **Upload the image:** Authenticate with a container registry such as Docker Hub and upload the tagged image.
11. **Prepare the cloud environment:** Select a cloud virtual machine or container hosting service capable of running Docker containers.
12. **Retrieve the image:** Log in to the cloud environment and download the required Docker image from the container registry.
13. **Deploy the application:** Start a container using the downloaded image and configure the necessary environment variables, storage, and port mappings.
14. **Configure network access:** Allow the application's required port through the cloud server's firewall or security settings.
15. **Access the deployed application:** Use the cloud server's public IP address or assigned domain to open the application from a web browser.
16. **Perform deployment testing:** Check the application's functionality, connectivity, logs, and response after deployment.
17. **Monitor the container:** Examine the container status and logs to identify any runtime errors or configuration problems.
18. **Complete deployment:** Once the application operates correctly, the Dockerized application is successfully running on the cloud infrastructure.



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Stepwise Screenshots with steps:

```
Command Prompt

C:\Users\Sheikhwasimuddin\Desktop>dir
Volume in drive C is OS
Volume Serial Number is A1B2-C3D4

Directory of C:\Users\Sheikhwasimuddin\Desktop

10-07-2025  10:15 AM    <DIR>          ..
10-07-2025  10:15 AM    <DIR>          hcs-lab
10-07-2025  10:15 AM                1,024  app.py
10-07-2025  10:15 AM                512  requirements.txt
10-07-2025  10:15 AM                321  Dockerfile
               3 File(s)                1,857 bytes
               2 Dir(s)  200,123,456,768 bytes free

C:\Users\Sheikhwasimuddin\Desktop>
```

```
Command Prompt

C:\Users\Sheikhwasimuddin\Desktop\hcs-lab>docker build -t sheikhwasimuddin/hcs-lab .
[+] Building 8.4s (8/8) FINISHED
=> [internal] load build definition from Dockerfile
=> transferring dockerfile: 312B
=> [internal] load metadata for docker.io/library/python:3.10-slim
=> [internal] load .dockerignore
=> [1/4] FROM docker.io/library/python:3.10-slim
=> [2/4] WORKDIR /app
=> [3/4] COPY requirements.txt .
=> [4/4] RUN pip install --no-cache-dir -r requirements.txt
=> exporting to image
=> => writing image sha256:8a7c2e3f9d1e4b6c9a2f3d8e7c1b9a0c2d3e4f5..
=> => naming to docker.io/sheikhwasimuddin/hcs-lab

C:\Users\Sheikhwasimuddin\Desktop\hcs-lab>
```



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```
Command Prompt

C:\Users\Sheikhwasimuddin\Desktop\hcs-lab> docker images

REPOSITORY          TAG          IMAGE ID       CREATED        SIZE
sheikhwasimuddin/hcs-lab  latest      8a7c2e3f9d1e   1 minute ago   125MB
python               3.10-slim   a1b2c3d4e5f6   2 weeks ago    124MB

C:\Users\Sheikhwasimuddin\Desktop\hcs-lab>
```

```
Command Prompt

C:\Users\Sheikhwasimuddin\Desktop\hcs-lab> docker run -d -p 5000:5000 --name
hcs-container sheikhwasimuddin/hcs-lab
f3d2a1b4c6e7d8f9a0b1c2d3e4f5a6c7b8d9e0f1

C:\Users\Sheikhwasimuddin\Desktop\hcs-lab> docker ps

CONTAINER ID   IMAGE                COMMAND                  CREATED        STATUS
f3d2a1b4c6e7   sheikhwasimuddin/hcs-lab  "python app.py"         5 seconds ago  Up 4 seconds
PORTS          NAMES
0.0.0.0:5000->5000/tcp  hcs-container

C:\Users\Sheikhwasimuddin\Desktop\hcs-lab>
```

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```
C:\Users\Sheikhwasimuddin\Desktop\hcs-lab> docker login
Authenticating with existing credentials...
Login Succeeded

C:\Users\Sheikhwasimuddin\Desktop\hcs-lab> docker push sheikhwasimuddin/hcs-lab:latest
The push refers to repository [docker.io/sheikhwasimuddin/hcs-lab]
a1b2c3d4e5f6: Pushed
c3d4e5f6a7b8: Pushed
d4e5f6a7b8c9: Pushed
e5f6a7b8c9d0: Pushed
latest: digest: sha256:9d1e4b6c3a7f8e2d1c9b0a5f6d3e7c2a1b4d8e9f0c...
size: 1250

C:\Users\Sheikhwasimuddin\Desktop\hcs-lab>
```

ubuntu@ip-172-31-22-45: ~

```
ubuntu@ip-172-31-22-45:~$ docker pull sheikhwasimuddin/hcs-lab:latest
latest: Pulling from sheikhwasimuddin/hcs-lab
Digest: sha256:9d1e4b6c3a7f8e2d1c9b0a5f6d3e7c2a1b4d8e9f0c...
Status: Downloaded newer image for sheikhwasimuddin/hcs-lab:latest
docker.io/sheikhwasimuddin/hcs-lab:latest

ubuntu@ip-172-31-22-45:~$ docker run -d -p 5000:5000 --name hcs-container
sheikhwasimuddin/hcs-lab:latest
a9b8c7d6e5f4a3b2c1d0e9f8a7b6c5d4e3f2a1b0

ubuntu@ip-172-31-22-45:~$ docker ps
CONTAINER ID   IMAGE                  COMMAND                  CREATED        STATUS
a9b8c7d6e5f4   sheikhwasimuddin/hcs-lab  "python app.py"         3 seconds ago  Up 2 seconds
PORTS          NAMES
0.0.0.0:5000->5000/tcp  hcs-container

ubuntu@ip-172-31-22-45:~$
```



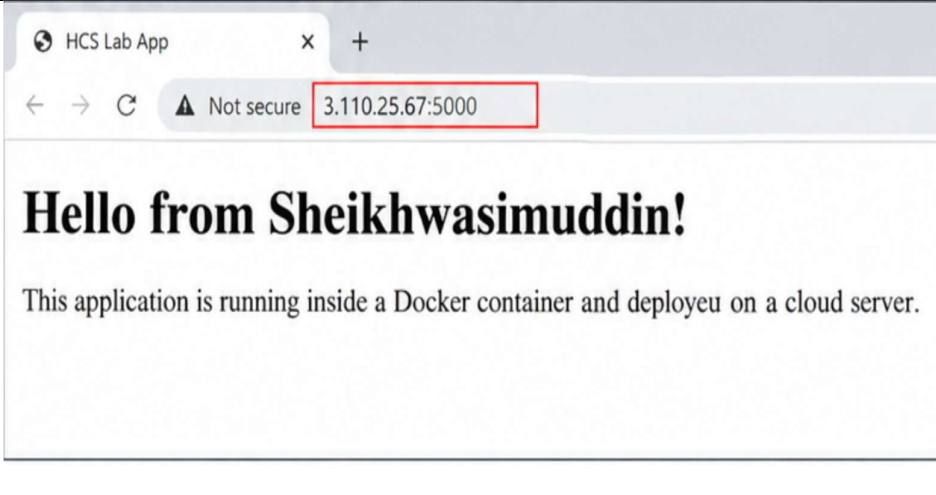
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Conclusion	Thus, the application was successfully containerized using Docker and deployed on a cloud server . The Docker image was created, the container was executed successfully, and the application was accessed through the cloud server. This demonstrates the use of Docker for consistent, portable, and efficient application deployment.
Date	21/08/2026